

1(a)	work done per unit mass	B1
	(work done) moving mass from infinity (to the point)	B1
1(b)(i)	gravitational potential energy = $(-)\frac{GMm}{r}$	C1
	$\Delta E_P = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 2.4 \times 10^3 \times [(6.4 \times 10^6)^{-1} - (1.2 \times 10^7)^{-1}]$	C1
	or	
	$\Delta \phi = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times [(6.4 \times 10^6)^{-1} - (1.2 \times 10^7)^{-1}]$	(C1)
	$\Delta E_P = m\Delta \phi$	(C1)
	$\Delta E_P = 7.0 \times 10^{10} \text{ J}$	A1
1(b)(ii)	$\frac{GMm}{r^2} = \frac{mv^2}{r}$	C1
	$v^2 = \frac{GM}{r}$ $= (6.67 \times 10^{-11} \times 6.0 \times 10^{24}) / (1.2 \times 10^7)$	C1
	$v = 5800 \text{ m s}^{-1}$	A1
1(c)	any one point from: - smaller gain in energy required if orbit is west to east - smaller change in velocity if orbit is west to east - smaller gain in energy if orbit is in same direction as Earth's rotation - smaller change in velocity if orbit is in same direction as Earth's rotation - satellite already moving west to east at launch - Earth's rotation is from west to east	B1